

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,collect(DISTINCT git_commit.sha) AS commitHashes\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n ,max(git_commit.sha) AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

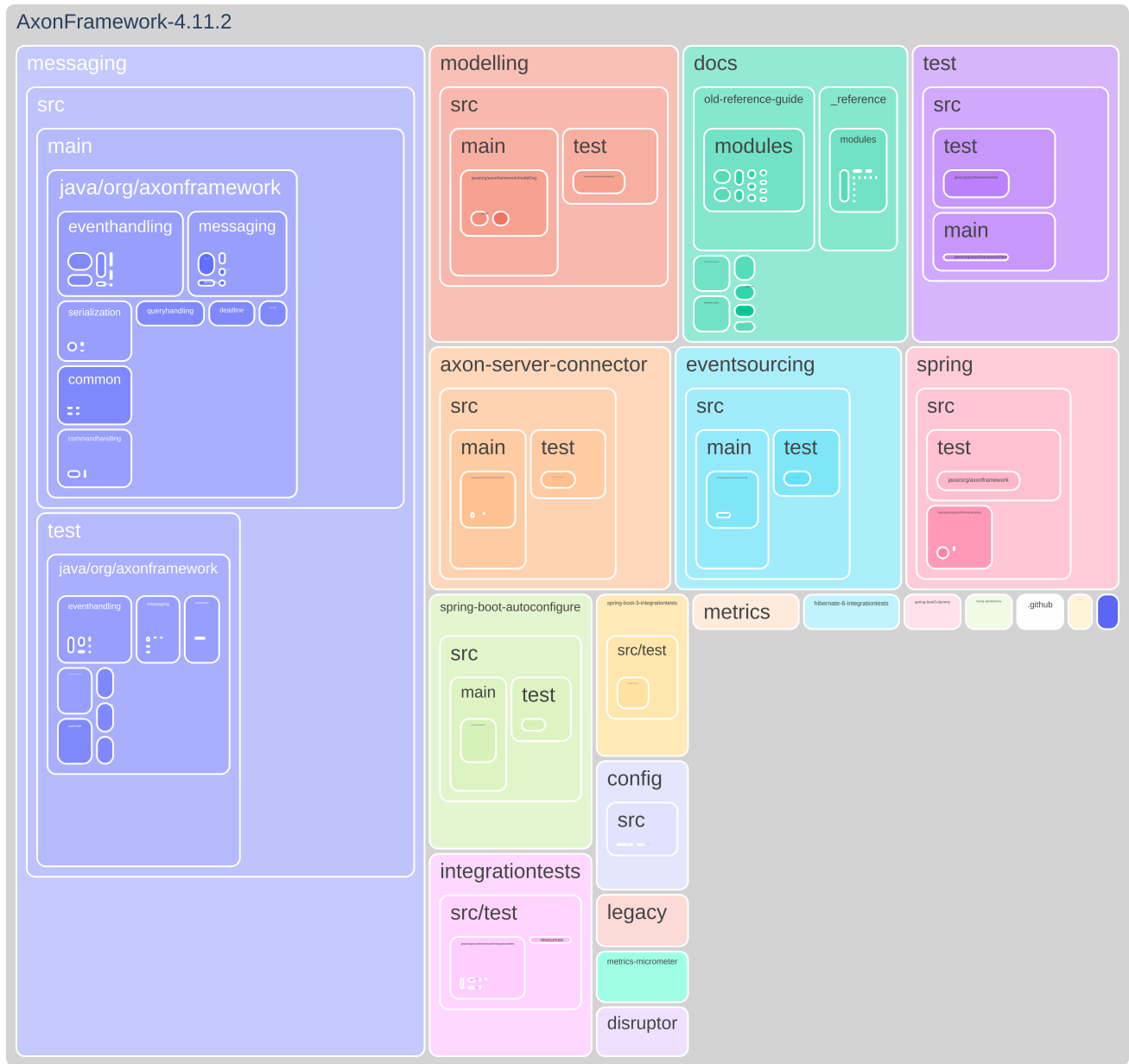
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	563.000000	563.000000	563.000000	563.000000	563.000000	563.000000
mean	24.492007	19.548845	283.804618	217.602131	1084.882771	310.534636
std	117.928146	21.254240	630.075836	137.280170	968.702100	373.632167
min	1.000000	2.000000	3.000000	18.000000	86.000000	17.000000
25%	1.000000	7.000000	27.000000	144.000000	326.000000	143.000000
50%	4.000000	12.000000	80.000000	286.000000	845.000000	285.000000
75%	10.000000	25.000000	262.000000	286.000000	1899.000000	285.000000
max	2250.000000	212.000000	9638.000000	2436.000000	5563.000000	2914.000000

	directoryPath	directoryParentPath
0	AxonFramework-4.11.2/assembly	AxonFramework-4.11.
1	AxonFramework-4.11.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.11.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat
2	AxonFramework-4.11.2/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.11.2/axon-server-connector/src/main/resources
3	AxonFramework-4.11.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.11.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event
4	AxonFramework-4.11.2/config/src/test/resources/META-INF	AxonFramework-4.11.2/config/src/test/resources
5	AxonFramework-4.11.2/coverage-report	AxonFramework-4.11.
6	AxonFramework-4.11.2/docs/_playbook/localLinks	AxonFramework-4.11.2/docs/_playbook
7	AxonFramework-4.11.2/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.2/docs/_reference/modules/ROOT
8	AxonFramework-4.11.2/docs/_reference/modules/ROOT/partials	AxonFramework-4.11.2/docs/_reference/modules/ROOT
9	AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector
10	AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.11.2/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.11.2/docs/_reference/modules/configuration/pages	AxonFramework-4.11.2/docs/_reference/modules/configuration
12	AxonFramework-4.11.2/docs/_reference/modules/configuration/partials	AxonFramework-4.11.2/docs/_reference/modules/configuration
13	AxonFramework-4.11.2/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.2/docs/_reference/modules/disruptor
14	AxonFramework-4.11.2/docs/_reference/modules/disruptor/partials	AxonFramework-4.11.2/docs/_reference/modules/disruptor
15	AxonFramework-4.11.2/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.2/docs/_reference/modules/eventsourcing
16	AxonFramework-4.11.2/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.11.2/docs/_reference/modules/eventsourcing
17	AxonFramework-4.11.2/docs/_reference/modules/legacy/pages	AxonFramework-4.11.2/docs/_reference/modules/legacy
18	AxonFramework-4.11.2/docs/_reference/modules/legacy/partials	AxonFramework-4.11.2/docs/_reference/modules/legacy
19	AxonFramework-4.11.2/docs/_reference/modules/metrics/pages	AxonFramework-4.11.2/docs/_reference/modules/metrics
20	AxonFramework-4.11.2/docs/_reference/modules/metrics/partials	AxonFramework-4.11.2/docs/_reference/modules/metrics
21	AxonFramework-4.11.2/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.2/docs/_reference/modules/micrometer
22	AxonFramework-4.11.2/docs/_reference/modules/micrometer/partials	AxonFramework-4.11.2/docs/_reference/modules/micrometer
23	AxonFramework-4.11.2/docs/_reference/modules/modeling/partials	AxonFramework-4.11.2/docs/_reference/modules/modeling
24	AxonFramework-4.11.2/docs/_reference/modules/spring/pages	AxonFramework-4.11.2/docs/_reference/modules/spring
25	AxonFramework-4.11.2/docs/_reference/modules/spring/partials	AxonFramework-4.11.2/docs/_reference/modules/spring
26	AxonFramework-4.11.2/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.2/docs/_reference/modules/springboot-starter
27	AxonFramework-4.11.2/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.11.2/docs/_reference/modules/springboot-starter
28	AxonFramework-4.11.2/docs/_reference/modules/test/pages	AxonFramework-4.11.2/docs/_reference/modules/test
29	AxonFramework-4.11.2/docs/_reference/modules/test/partials	AxonFramework-4.11.2/docs/_reference/modules/test

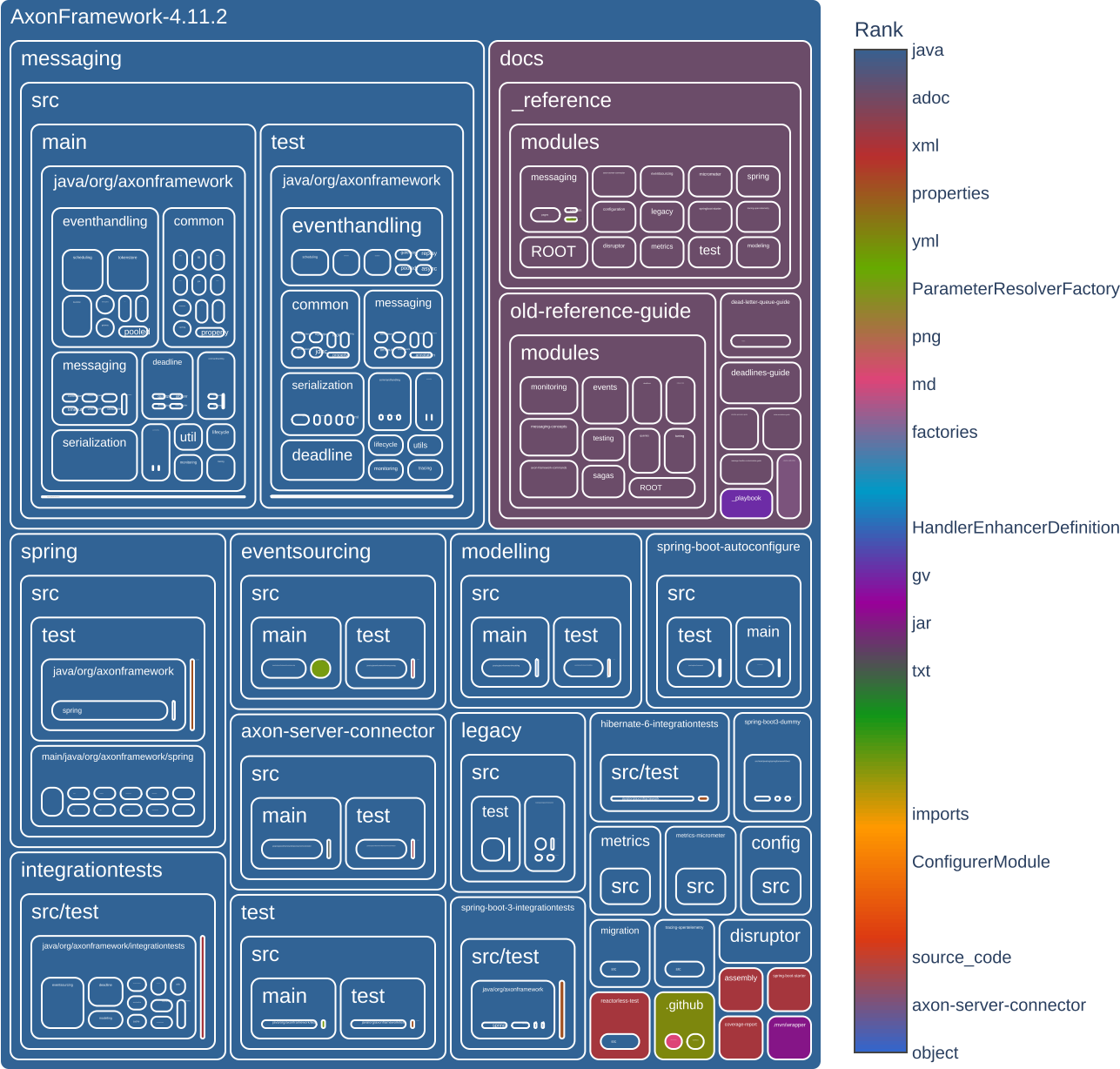
Number of files per directory

Directories and their file count



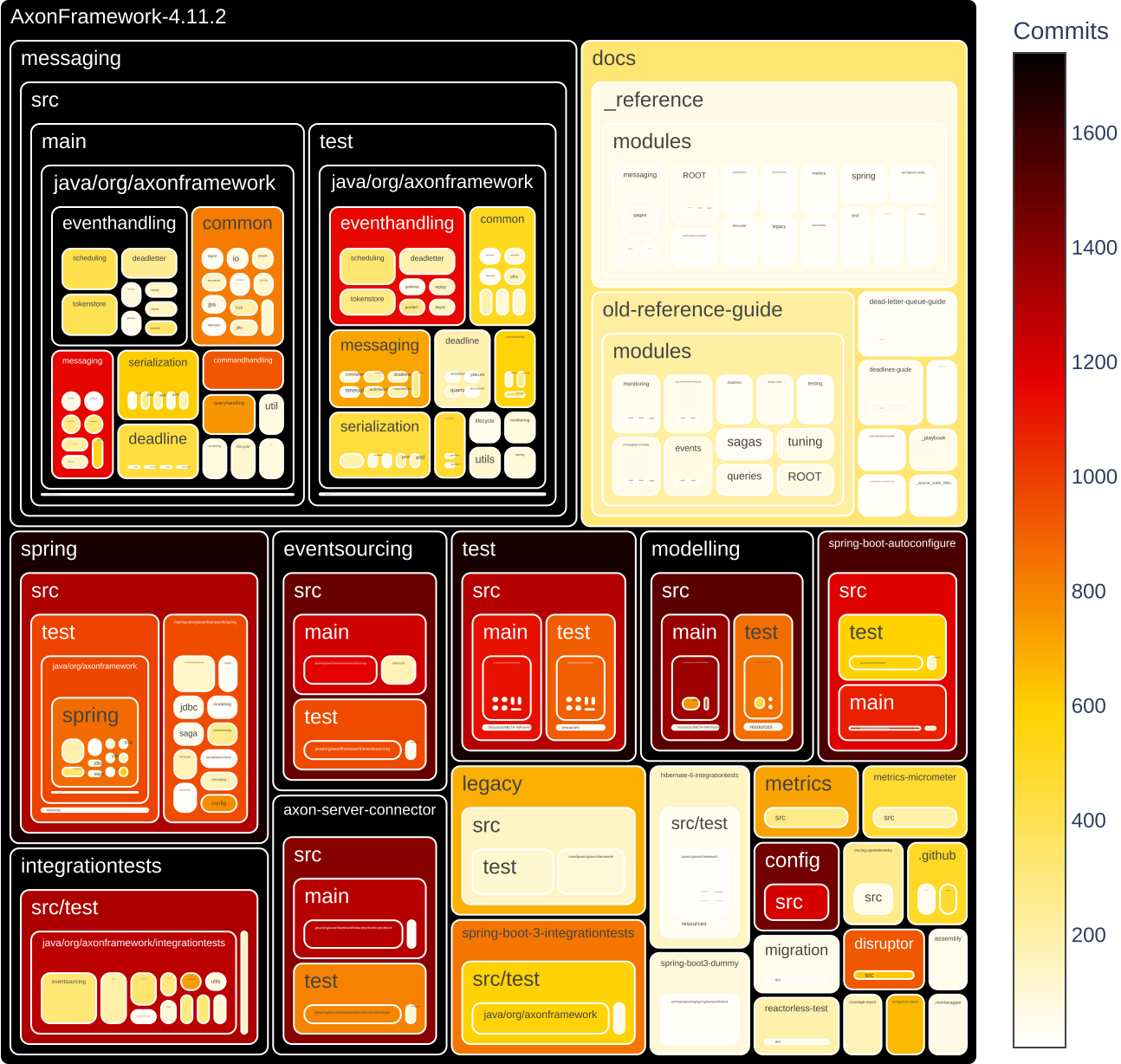
Most frequent file extension per directory

Most frequent file extension per directory



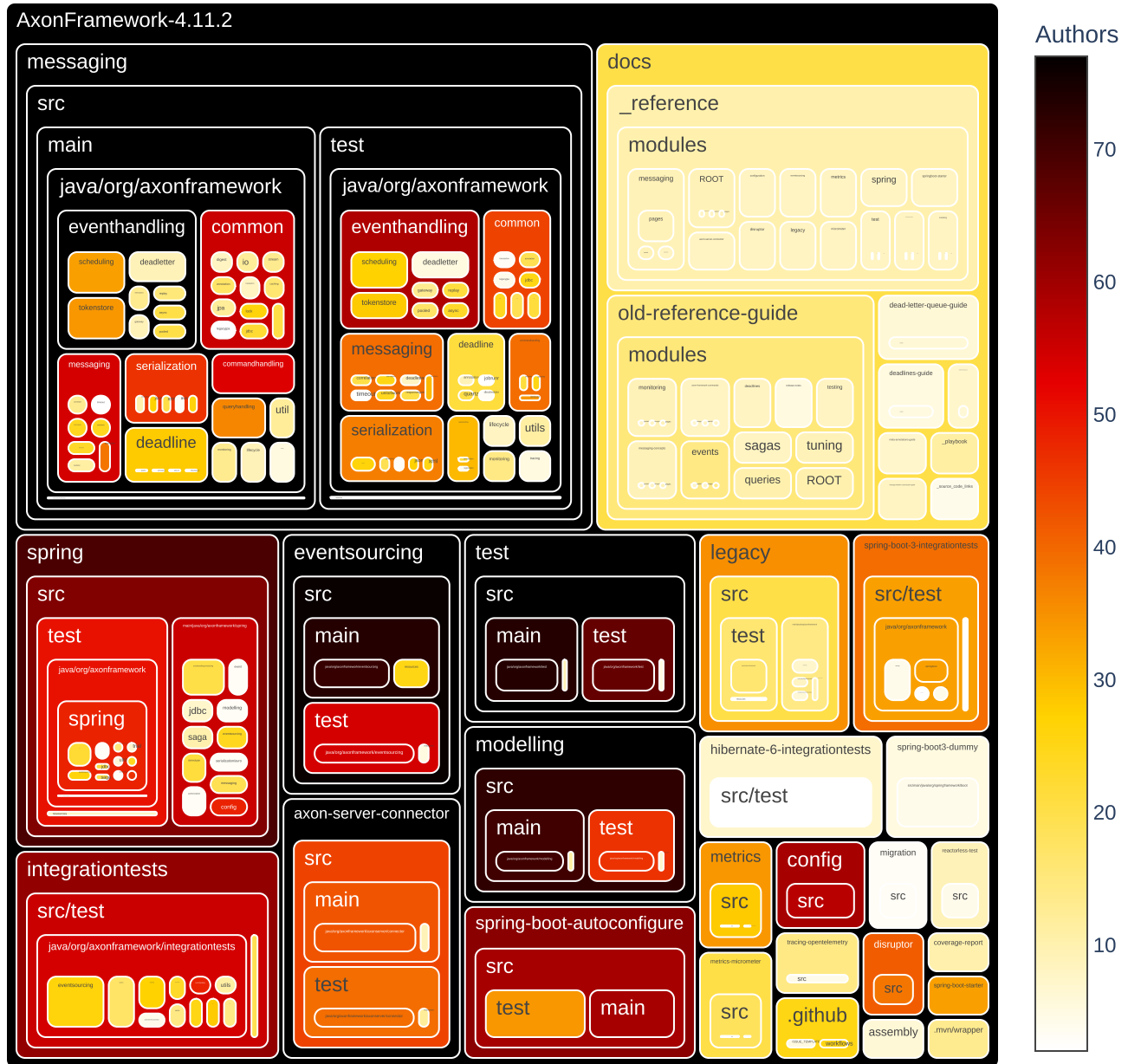
Number of commits per directory

Number of git commits



Number of distinct authors per directory

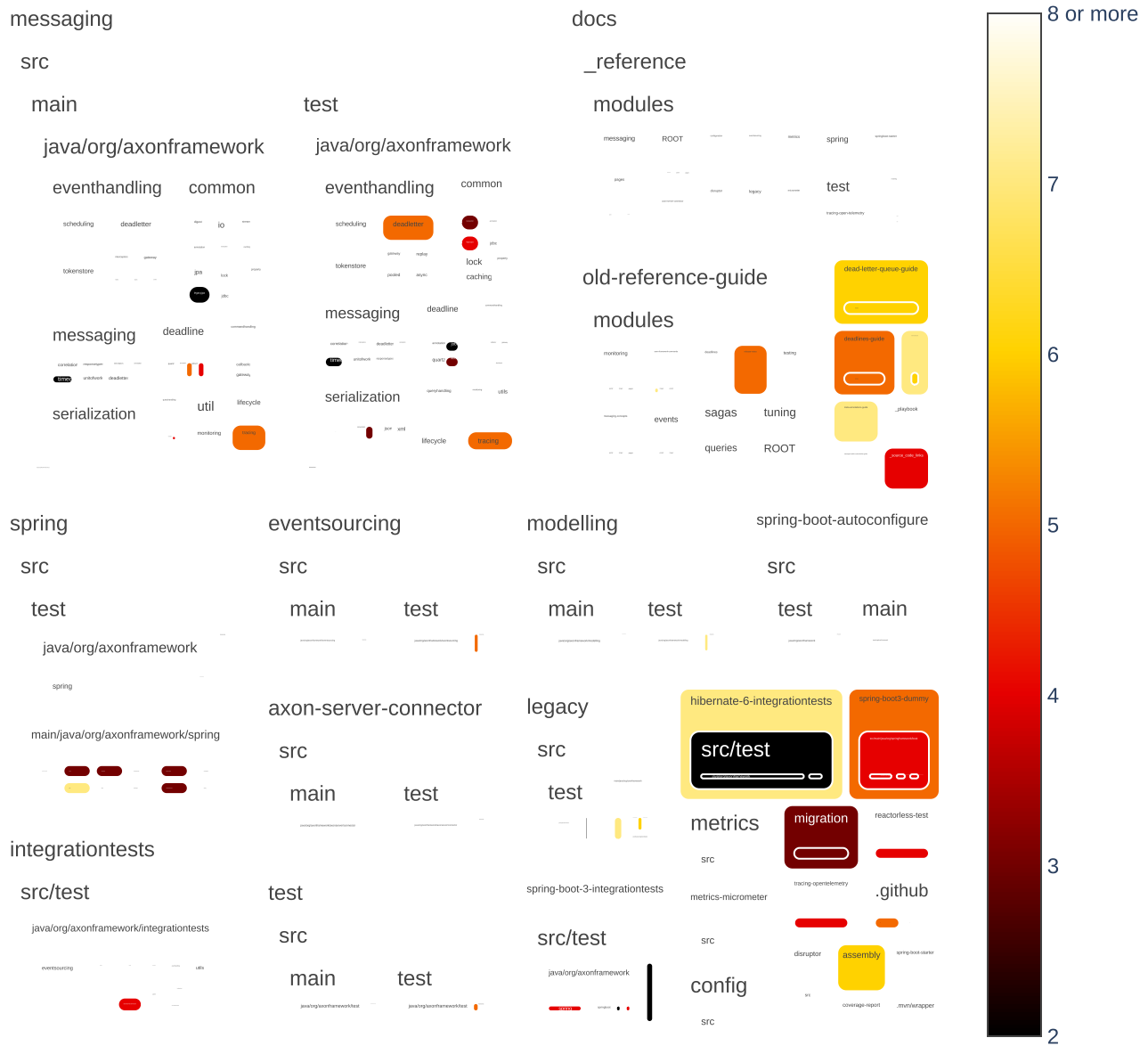
Number of distinct commit authors



Directories with very few different authors

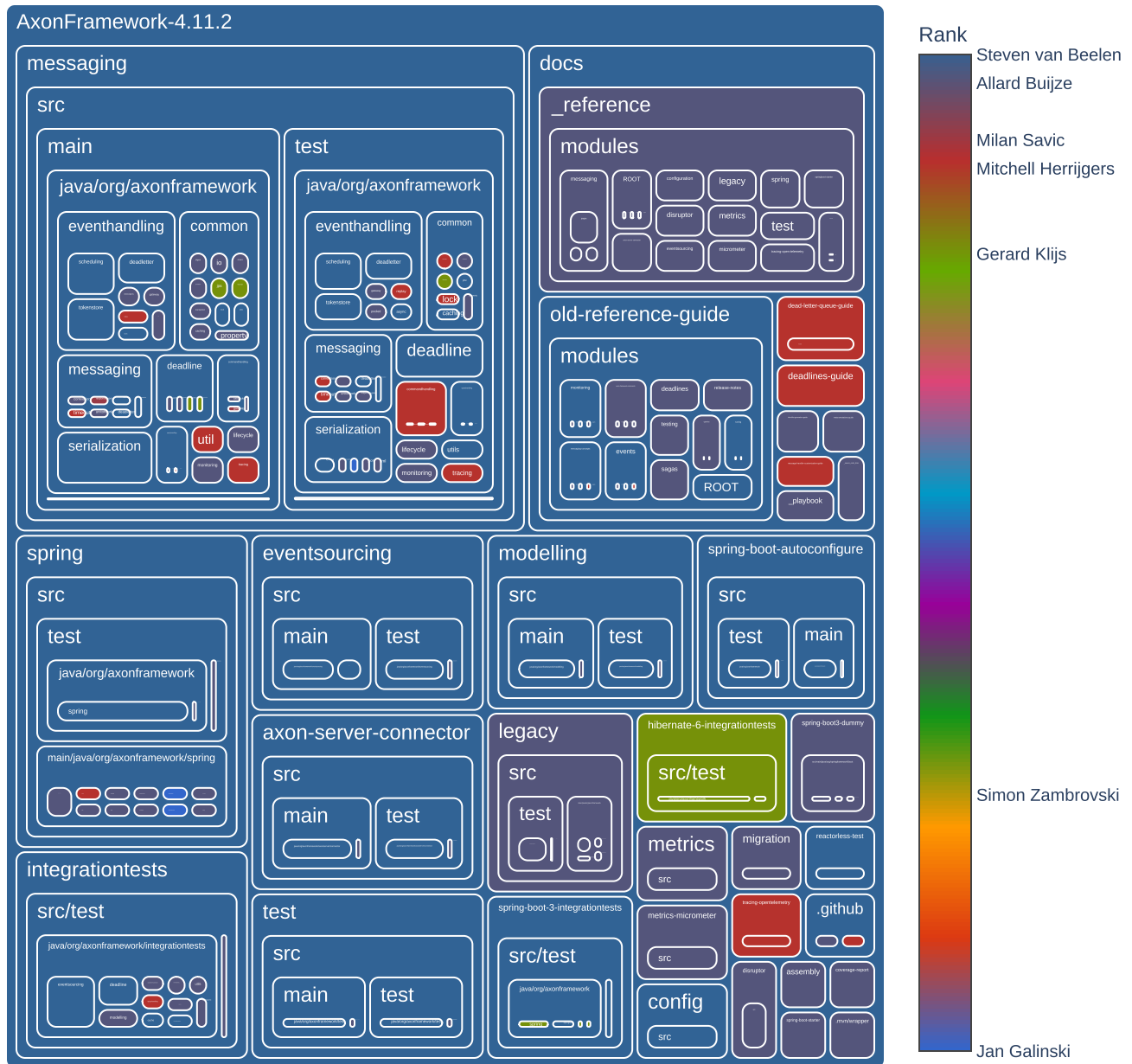
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.2



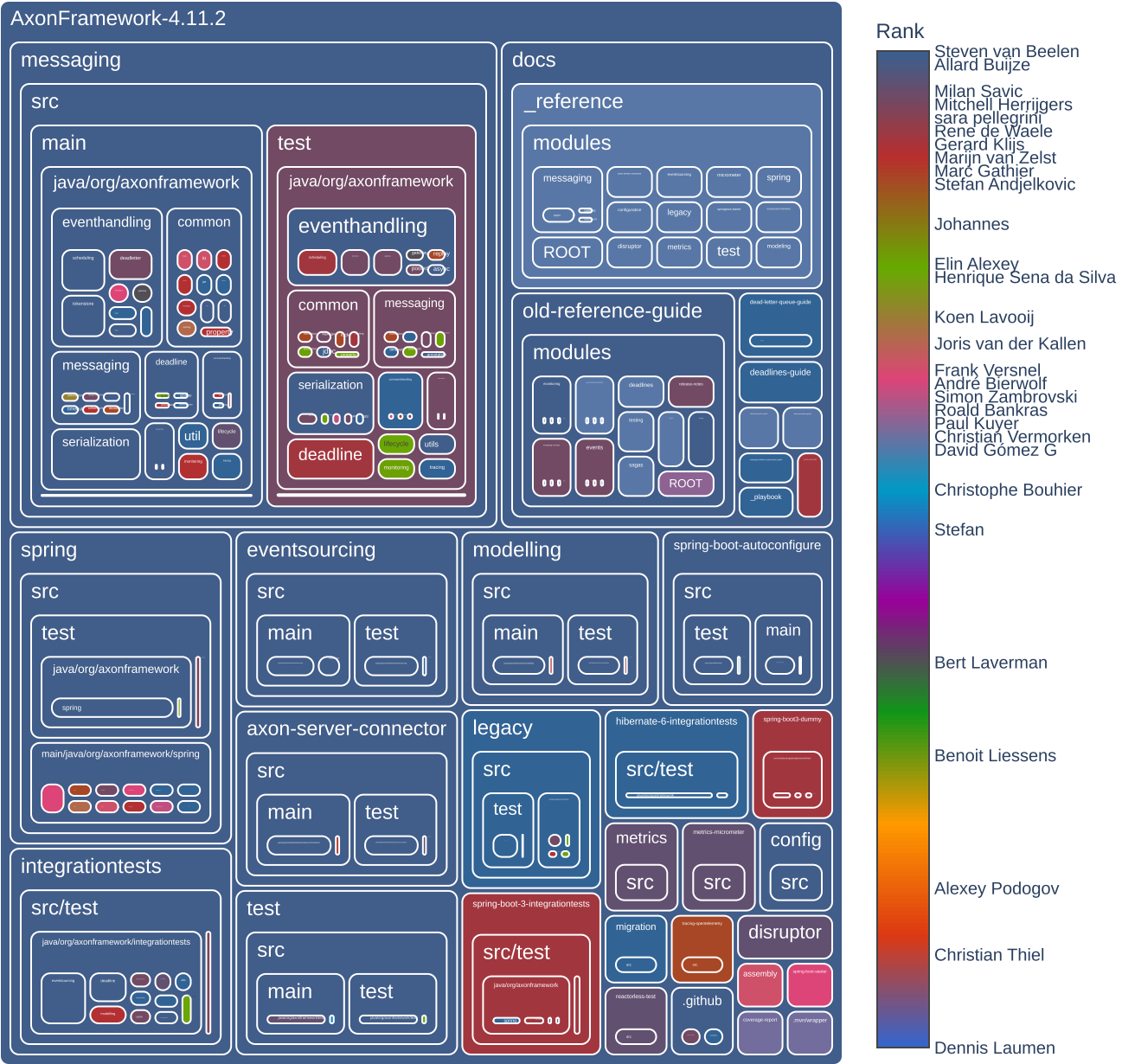
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

AxonFramework-4.11.2

messaging

src

main

java/org/axonframework



test

java/org/axonframework



docs

_reference

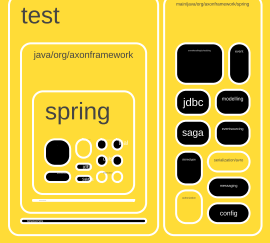


old-reference-guide



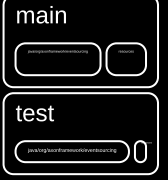
spring

src



eventsourcing

src



test

src



modelling

src



spring-boot-autoconfigure

src



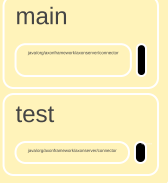
integrationtests

src/test



axon-server-connector

src



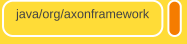
legacy

src



spring-boot-3-integrationtests

src/test



metrics

src/test



spring-boot-dummy



metrics-micrometer

src



config

src



migration

reactorless-test

src

disruptor

assembly

spring-boot-3-integrationtests

src/test

java/org/axonframework

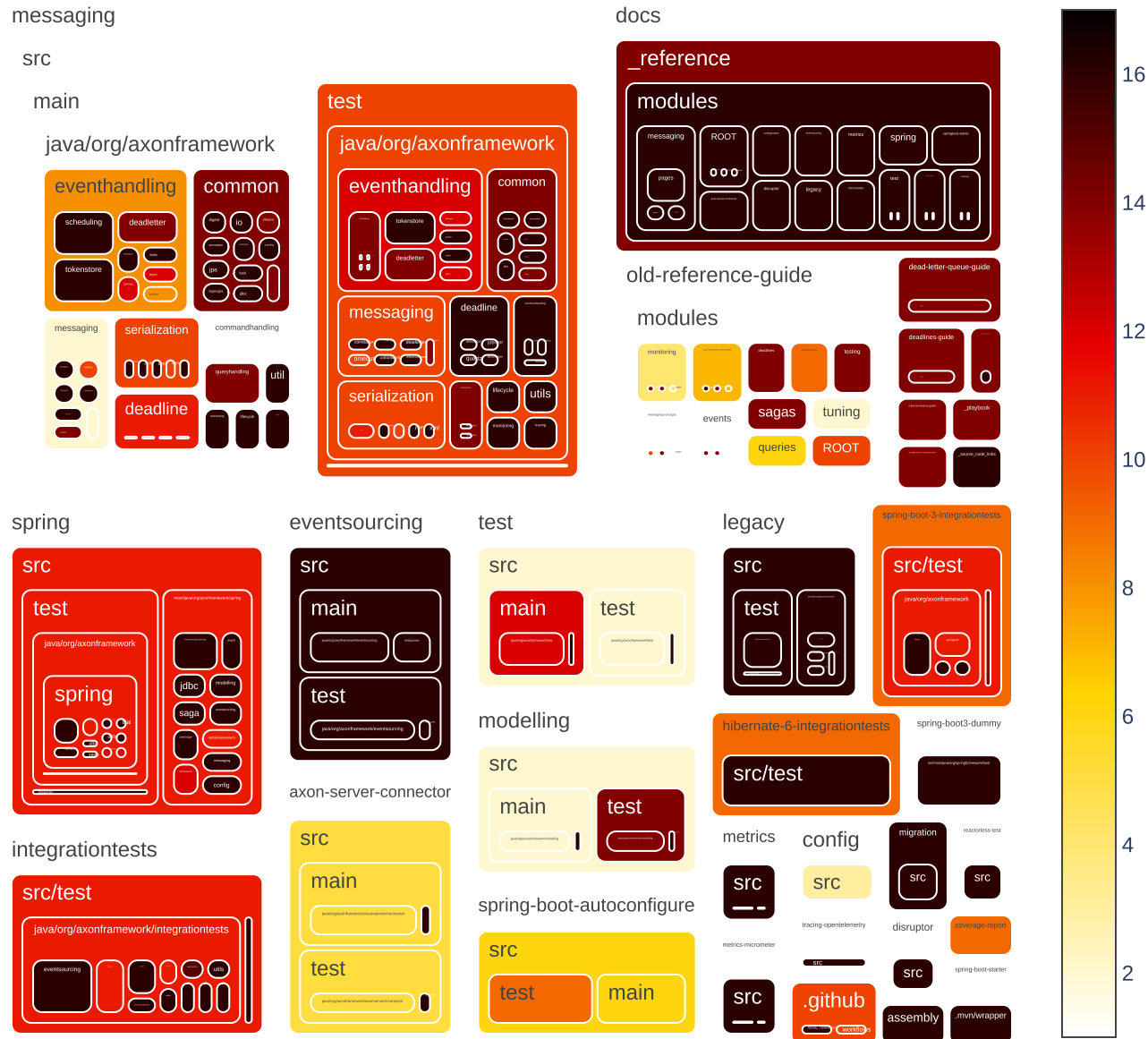
Days



Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.11.2



Days since last file creation per directory

Days since last file creation

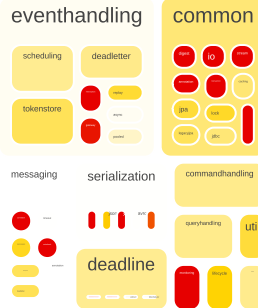
AxonFramework-4.11.2

messaging

src

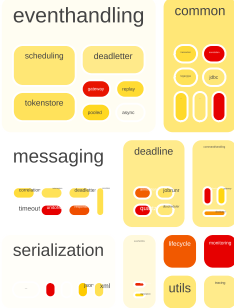
main

java/org/axonframework

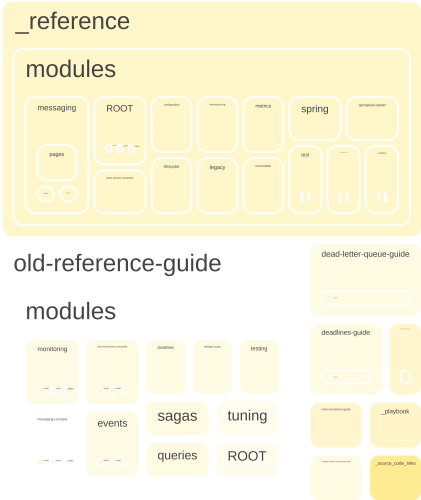


test

java/org/axonframework

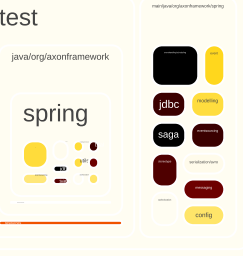


docs



spring

src



integrationtests

src/test



eventsourcing

src



axon-server-connector

src

main

test

java/org/axonframework

test

test

src



legacy

src

test

spring-boot-3-integrationtests

src/test

java/org/axonframework

modelling

src



spring-boot-autoconfigure

src

test

main

test

main

test

main

test

main

test

main

test

Days

3500

3000

2500

2000

1500

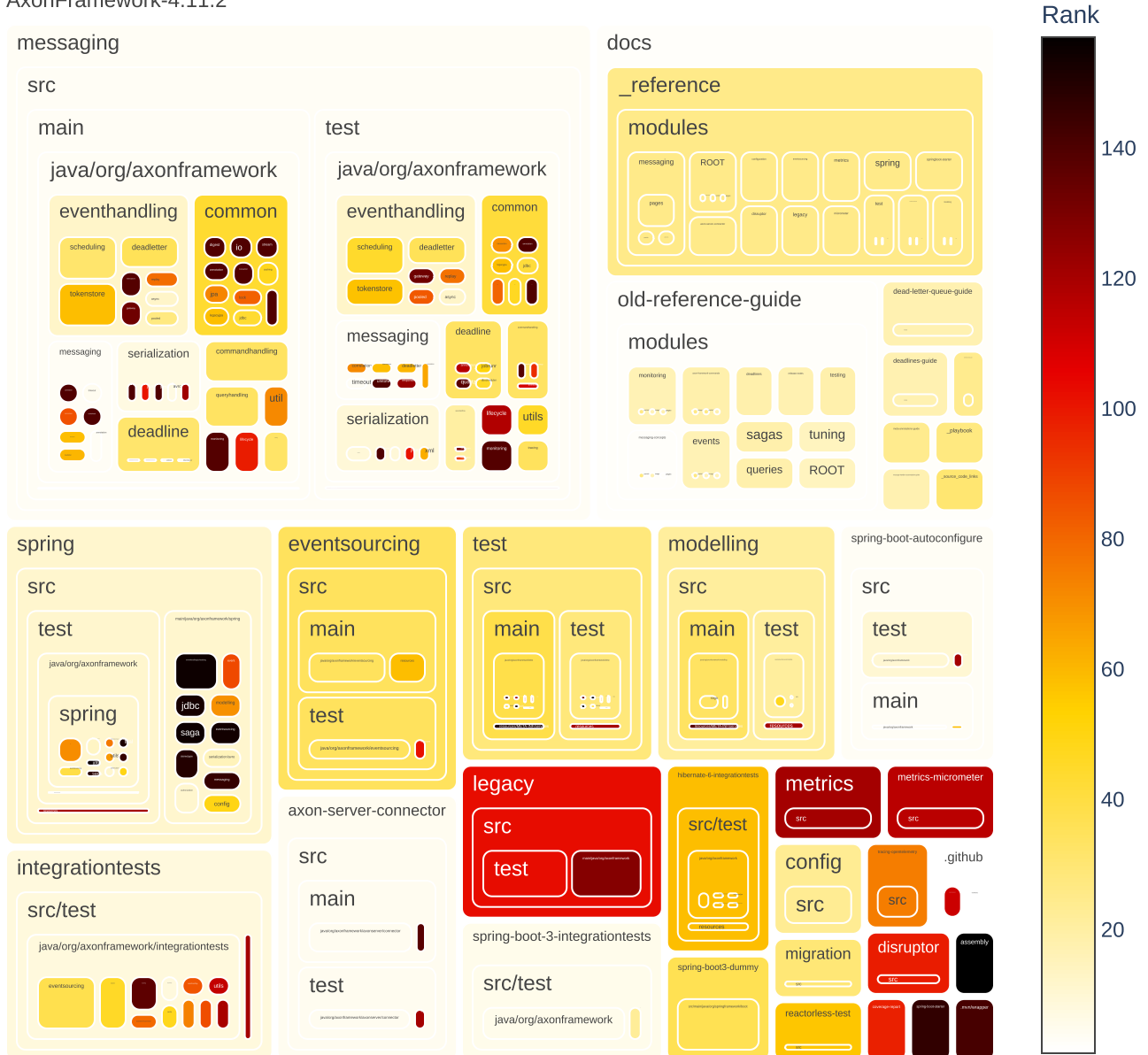
1000

500

Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.2



Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.2

messaging

src

main

java/org/axonframework



test

java/org/axonframework



docs

_reference

modules



old-reference-guide

modules



Days

2000

1500

1000

500

spring

src

test



eventsourcing

src

main

test



test

src

main

test



modelling

src

main

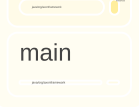
test



spring-boot-autoconfigure

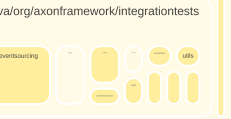
src

test



integrationtests

src/test

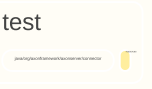


axon-server-connector

src

main

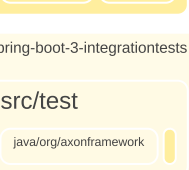
test



legacy

src

test



hibernate-6-integrationtests

src/test

src/test

src/test

src/test

src/test

src/test

metrics

src

src

src

src

src

src

metrics-micrometer

src

src

src

src

src

src

Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-4.11.2

messaging

src

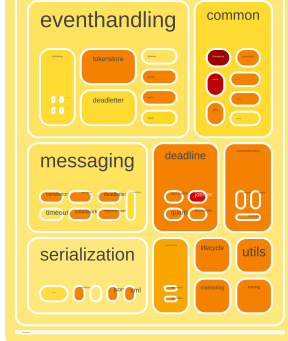
main

java/org/axonframework



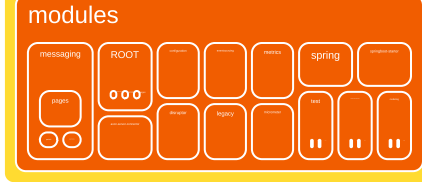
test

java/org/axonframework



docs

_reference



old-reference-guide

modules

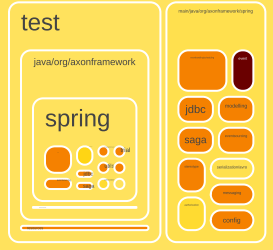


dead-letter-queue-guide



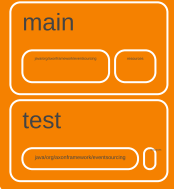
spring

src



eventsourcing

src



test

src



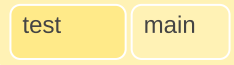
modelling

src



spring-boot-autoconfigure

src



legacy

src



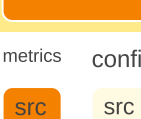
spring-boot-3-integrationtests

src/test



hibernate-6-integrationtests

src/test



spring-boot3-dummy



integrationtests

src/test



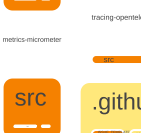
src

main



metrics

src



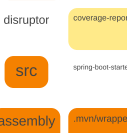
config

src



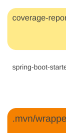
migration

src



reactorless-test

src



coverage-report

spring-boot-starter

assembly

.myn/wrapper

Rank

50

40

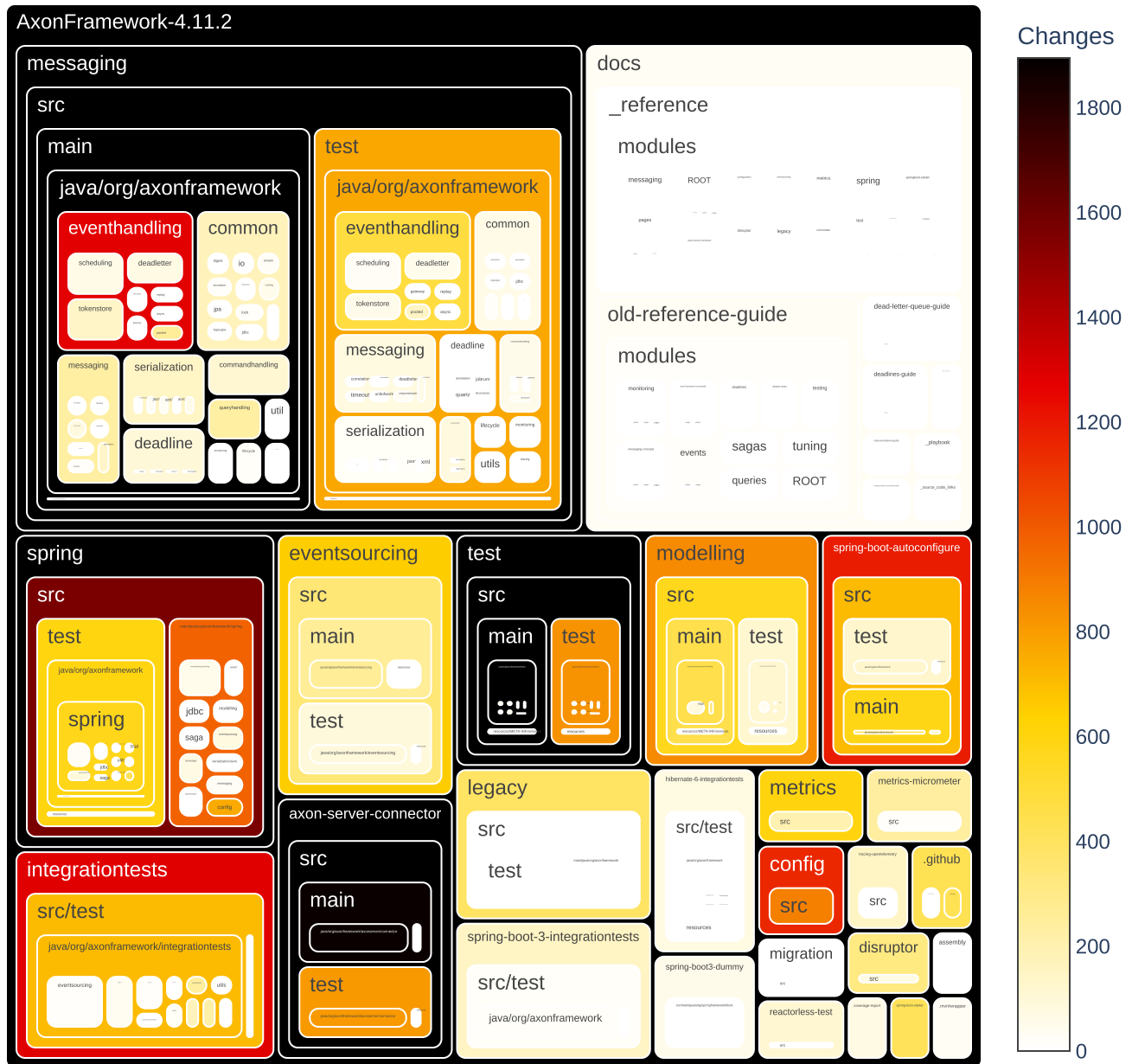
30

20

10

File changed frequently with other files

Pairwise file changes



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

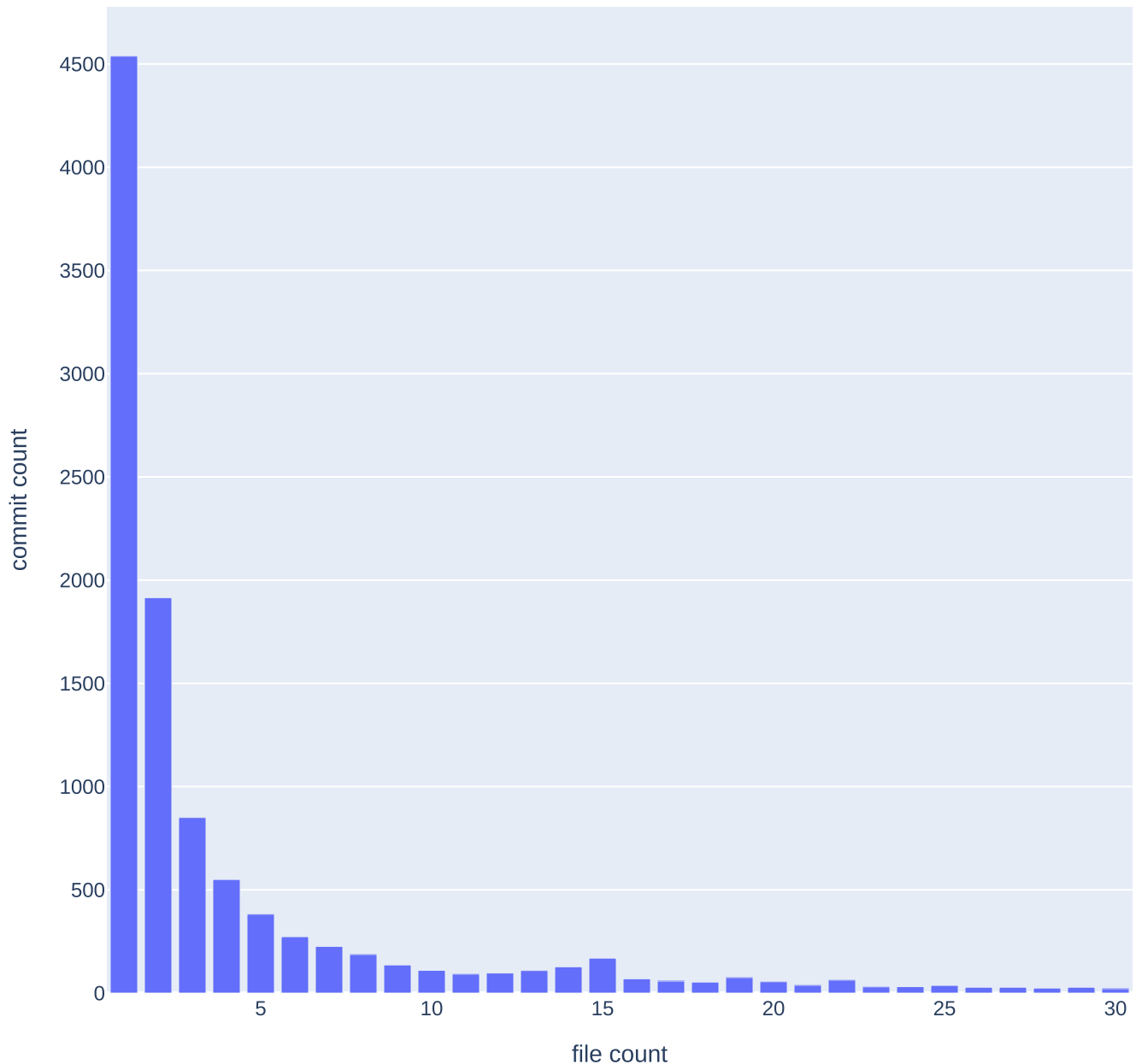
```
Sum of commits that changed more than 30 files (each) = 961
Max changed files with one commit = 2178
```

	filesPerCommit	commitCount
count	333.000000	333.000000
mean	344.075075	34.114114
std	440.375494	276.321358
min	1.000000	1.000000
25%	84.000000	1.000000
50%	186.000000	2.000000
75%	428.000000	5.000000
max	2178.000000	4538.000000

	filesPerCommit	commitCount
0	1	4538
1	2	1915
2	3	850
3	4	550
4	5	382
5	6	273
6	7	226
7	8	187
8	9	136
9	10	110
10	11	92
11	12	97
12	13	109
13	14	127
14	15	169
15	16	69
16	17	58
17	18	53
18	19	75
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	24
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.

- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: cardinality)} {position: line: 9, column: 48, offset: 576} for query:
"// List pair of files that were changed together and that have a declared dependency between each other.\n\nMATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\nMATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n//De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n//WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\nWITH firstCodeFile.fileName AS firstFileName\n,secondCodeFile.fileName AS secondFileName\n,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n,pairwiseChange.commitCount AS commitCount\n,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\nRETURN dependencyWeight\n,commitCount\n,fileDistanceAsFewestChangeDirectoryCommands\n// ,count(*) AS occurrences\n// ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\nORDER BY dependencyWeight, commitCount\n\n// MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n// MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n// WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n// RETURN firstCodeFile.fileName AS firstFileName\n// ,secondCodeFile.fileName AS secondFileName\n// ,dependency.weight AS dependencyWeight\n// ,pairwiseChange.commitCount AS commitCount\n// ORDER BY dependencyWeight, commitCount\n\n// MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n// WITH count(DISTINCT relation) AS relatedFilesCount\n// ,collect(DISTINCT relation) AS relations\n// UNWIND relations AS relation\n// WITH relatedFilesCount\n// ,coalesce(relation.commitCount, 0) AS commitCount\n// ,coalesce(relation.weight, 0) AS dependencyWeight\n// ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n// RETURN dependencyWeight\n// ,commitCount\n// ,fileDistanceAsFewestChangeDirectoryCommands\n// ORDER BY dependencyWeight, commitCount"
```

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
0	1	12	0
1	1	12	0
2	1	12	0
3	1	12	2
4	1	12	0
5	1	12	4
6	1	12	4
7	1	12	0
8	1	12	0
9	1	12	4
10	1	13	0
11	1	13	4
12	1	13	0
13	1	13	4
14	1	14	2
15	1	14	4
16	1	15	2
17	1	15	0
18	1	15	0
19	1	16	0

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	166.000000	166.000000	166.000000
mean	9.566265	25.253012	1.373494
std	19.257863	16.677579	1.570260
min	1.000000	12.000000	0.000000
25%	1.250000	14.250000	0.000000
50%	3.000000	19.000000	0.000000
75%	10.000000	28.000000	2.000000
max	154.000000	95.000000	4.000000

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.204462	-0.143721
commitCount	0.204462	1.000000	-0.143874
fileDistanceAsFewestChangeDirectoryCommands	-0.143721	-0.143874	1.000000

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	1.000000	0.212543	-0.089280
commitCount	0.212543	1.000000	-0.132659
fileDistanceAsFewestChangeDirectoryCommands	-0.089280	-0.132659	1.000000

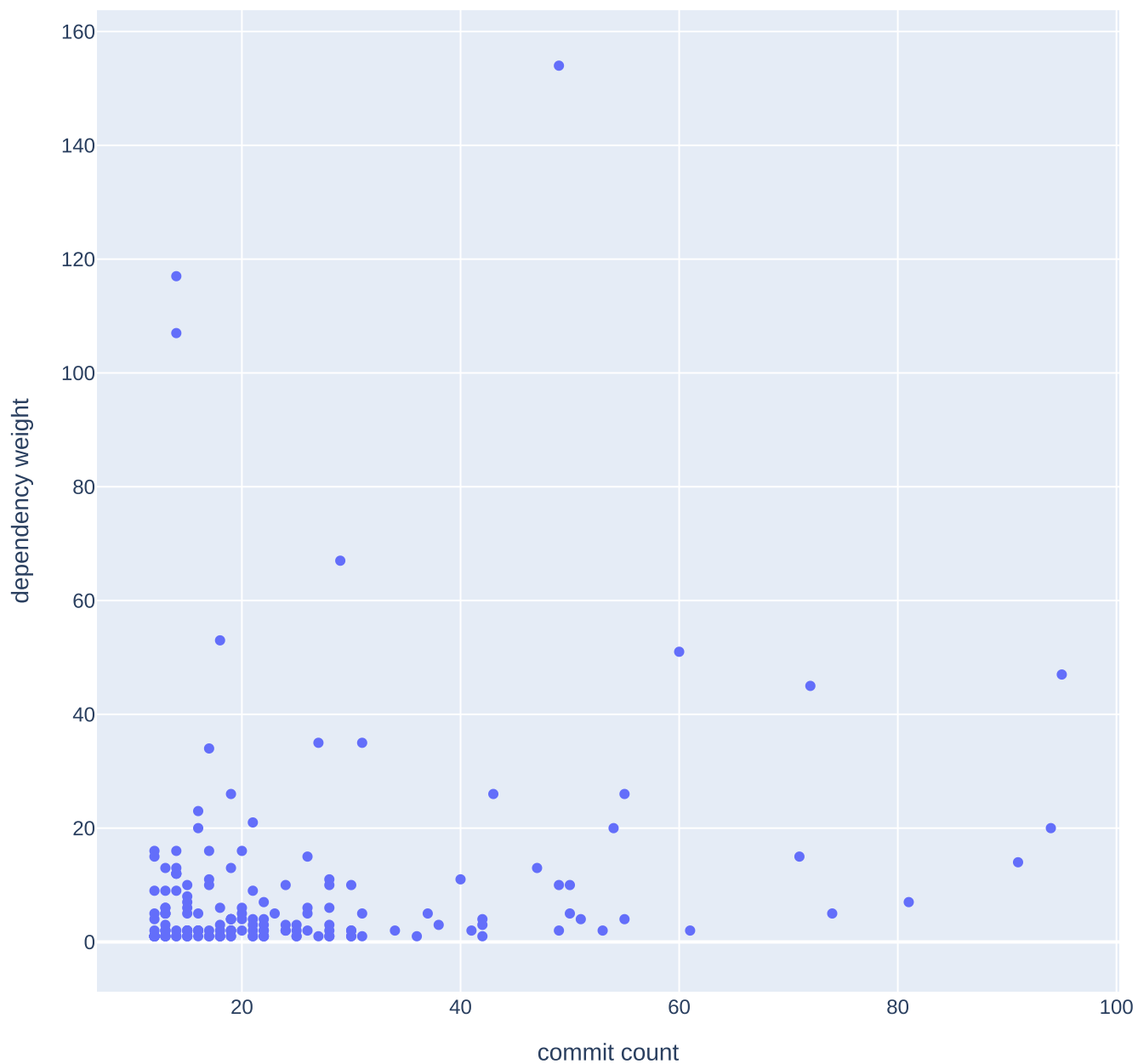
'Pearson Correlation with p-value for commitCount and dependencyWeight'

PearsonRResult(statistic=0.20446197739019414, pvalue=0.008232632527754543)

'Spearman Correlation with p-value for commitCount and dependencyWeight'

SignificanceResult(statistic=0.21254323842799402, pvalue=0.005973735857352672)

Pairwise changed files: Number of changes (commitCount) vs. dependency weight



WordCloud of git authors

	word	frequency
0	Steven van Beelen	3402
1	Allard Buijze	2722
2	smcvb	822
3	Milan Savic	449
4	Mitchell Herrijgers	313
5	Gerard Klijs	246
6	Marc Gathier	231
7	sara pellegrini	163
8	Rene de Waele	121
9	Jетро Coenradie	117

Wordcloud of git authors

